Mittuniversitetet
MID SWEDEN UNIVERSITY

Climate Change Impact Reduction Strategies for Promotional Hats at the Biathlon World Cup in Östersund

A Scenario-Based Life Cycle Assessment

Greta Stankute

Independent Project
Main field of study: Environmental Science
Credits: 15 hp
Semester/year: VT/2026
Supervisor: Judith Waller
Examiner: Erik Grönlund
Course code/registration number: MX045G
Programme: Ecotechnology (NEKOG), 180 hp

At Mid Sweden University, it is possible to publish the thesis in full text in DiVA (see appendix for publishing conditions). The publication is open access, which means that the work will be freely available to read and download online. This increases the dissemination and visibility of the degree project.

Open access is becoming the norm for disseminating scientific information online. Mid Sweden University recommends both researchers and students to publish their work open access.

I/we allow publishing in full text (free available online, open access):

☒ Yes, I/we agree to the terms of publication.

☐ No, I/we do not accept that my independent work is published in the public interface in DiVA (only archiving in DiVA).

Östersund, 2026-06-05 .....

Location and date

Ecotechnology (NEKOG).....

Programme/Course

Greta Stankute.....

Name (all authors names)

2003 .....

Year of birth (all authors year of birth)

Abstract

Branded textiles distributed at sporting events represent an understudied source of environmental impact, characterised by large procurement volumes, short distribution windows, and limited reuse. This study quantifies the climate change impact of hats distributed at the Biathlon World Cup in Östersund and evaluates three procurement-level intervention strategies using comparative scenario-based life cycle assessment (LCA). The functional unit is one event edition of hat provisioning, covering 1,530 acrylic hats distributed across volunteer, giveaway, and merchandise channels. Modelling was conducted in SimaPro using the ecoinvent 3 database and the ReCiPe 2016 Midpoint (H) method, with climate change reported in kg CO_2e.

The baseline system generates 1,670 kg CO_2e per edition. A formalised volunteer bring-back scheme at a 30% return rate reduces this by 19% (1,350 kg CO_2e). Reducing public giveaways from 30 to 10 units yields a 2% reduction, reflecting the channel's small share of total procurement volume. The combined strategy achieves a 21% reduction. Sensitivity analysis confirms that the reuse rate is the dominant source of uncertainty, while end-of-life treatment and unit weight assumptions are secondary and do not alter scenario rankings.

The study concludes that procurement volume reduction through reuse is structurally more effective than material substitution or end-of-life management for this product system. Key limitations include the use of a polyester proxy for acrylic raw material, a single reported impact category, and the absence of empirically derived reuse rates. The functional unit and scenario framework developed here offer a replicable template for LCA of event merchandise distribution at other venues and scales.

Keywords: life cycle assessment; circular procurement; textile accessories; sporting events; reuse; sensitivity analysis

Table of Contents

Abstract *3*

List of Abbreviations *6*

1. Introduction *7*

1.1 The Environmental Footprint of the Textile Industry **7**

1.2 Acrylic as a Material **8**

1.3 Textile Distribution at Sporting Events **8**

1.4 Service Life and Reuse **9**

1.5 Life Cycle Assessment: Method and Application **10**

1.6 Aim and Research Question **11**

2. Method *11*

2.1 LCA Framework and Software **11**

2.2 Functional Unit and System Boundary **12**

2.3 Scenarios **12**

2.4 Life Cycle Inventory **13**

2.5 Sensitivity Analysis **15**

3. Results *15*

3.1 Life cycle inventory **15**

3.2 Baseline System (S0) **16**

3.3 Scenario Comparison **16**

3.4 Sensitivity Analysis **18**

3.4.1 Reuse rate 18

3.4.2 End-of-life treatment 19

3.4.3 Hat unit weight 19

3.4.4 Manufacturing electricity input 19

4. Discussion *20*

4.1 Which Lever Matters and Why **20**

4.2 Comparison with Existing Literature **21**

4.3 Limitations and Uncertainties **22**

4.4 Broader Implications **23**

4.5 Practical Recommendations for the Organiser **24**

4.6 Conclusion **26**

References *27*

Terms of publishing *28*

List of Abbreviations

Abbreviation	Definition
CN	China (ecoinvent geography)
CO_2e	Carbon dioxide equivalent
Dtex	Decitex
GLO	Global (ecoinvent geography)
IBU	International Biathlon Union
ISO	International Organization for Standardization
kWh	Kilowatt-hour
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
LCIA	Life Cycle Impact Assessment
ReCiPe	Impact assessment method (Rijksinstituut voor Volksgezondheid en Milieu)
RoW	Rest of World (ecoinvent geography)
S0	Baseline scenario, full procurement each edition
S1	Reuse scenario, 30% volunteer bring-back
S2	Reduced giveaways scenario
S3	Combined scenario (S1 + S2)
tkm	Tonne-kilometre

1. Introduction

1.1 The Environmental Footprint of the Textile Industry

The textile industry is one of the most resource-intensive sectors in the world. It is responsible for more than 1.7 billion tonnes of CO_2e emissions per year and continues to grow as fast-fashion consumption increases (Niinimäki et al., 2020). Beyond climate change, the sector is responsible for around 20% of industrial water pollution from textile treatment and dyeing, and contributes roughly 190,000 tonnes of primary microplastics to the oceans annually (Niinimäki et al., 2020). These figures reflect a production model built on high volumes, short product lifespans, and disposal as the default end-of-life pathway.

A large share of these impacts is tied to how fibres are made. Synthetic materials such as polyester, polyamide, and acrylic are derived from fossil fuel resources and require energy-intensive processing at every stage: polymerisation, spinning, knitting, dyeing, and finishing. Research comparing the life cycle performance of different fibre types has found that environmental outcomes vary significantly depending on yarn thickness, manufacturing processes, and end-of-life treatment (van der Velden et al., 2014). General claims about which fibre is environmentally preferable are therefore often misleading without a specific product context, which is one reason LCA studies in this sector need to be carefully scoped.

In response to these challenges, life cycle assessment (LCA) has become the principal method for measuring and comparing environmental performance in textiles. A systematic review of peer-reviewed textile LCAs from 2010 to 2024 found that the field has grown considerably in sophistication, but that key limitations remain: system boundaries are often set inconsistently, functional units are difficult to compare across studies, and impact categories such as biodiversity, water toxicity, and microplastics remain underrepresented (Dhiwar & Bedarkar, 2025). This study focuses on climate change as its primary impact category, consistent with standard practice in textile LCA, and acknowledges that additional categories were not fully assessed due to data constraints - a limitation addressed further in Section 4.3.

1.2 Acrylic as a Material

The hats distributed at the Biathlon World Cup in Östersund are 100% acrylic, a synthetic fibre whose production characteristics and end-of-life limitations shape the environmental logic of this study (J. Engström, personal communication, 27 February 2026). Acrylic is produced through the polymerisation of acrylonitrile, a petrochemical monomer, making it fully dependent on fossil resources. It does not biodegrade meaningfully under normal conditions, and unlike polyester, it has no well-established recycling pathway at commercial scale. Mechanical recycling is difficult due to the fibre's chemical structure, and chemical recycling options for acrylic remain largely undeveloped (Sandin & Peters, 2018). In practice, acrylic products that reach end of life are mostly landfilled or incinerated.

A benchmarking LCA study by van der Velden et al. (2014), published in the International Journal of Life Cycle Assessment, compared acrylic, polyester, nylon, elastane, and cotton across several environmental indicators on a cradle-to-grave basis. Acrylic and polyester showed the lowest impacts per kilogram of finished textile. Importantly, for synthetic fibres, the production phase - particularly spinning and knitting - is the dominant life cycle stage, not the use phase. This differs from natural fibres such as cotton, where washing during use tends to dominate. For acrylic, producing fewer items or extending the service life of existing items is therefore the most direct way to reduce environmental impact.

At the scale of a sporting event distributing over 1,500 hats per edition, this finding has direct practical relevance. Each unit manufactured carries a fixed production burden, and the environmental cost of that burden is determined by how many times the hat is actually used before reaching end of life.

1.3 Textile Distribution at Sporting Events

Distributing branded textiles to volunteers, staff, and spectators is standard practice at large sporting events. The items serve practical functions such as identifying personnel and providing event clothing, but also function as promotional merchandise tied to the event and its commercial partners. At the Biathlon World Cup in Östersund, hats are distributed through three channels: 1,000 go to volunteers as part of their event equipment, 30 are given away to the public at entertainment

stands, and 500 are sold through the merchandise shop (J. Engström, personal communication, 27 February 2026).

Each distribution channel implies a different relationship between the recipient and the product. A hat purchased from a merchandise shop is likely to be used repeatedly. A hat received as a free giveaway at an entertainment stand may be worn once, stored, or never used. This variation in use intensity has a bearing on environmental performance in per-use terms, though the absolute production burden is identical regardless of how the hat was obtained.

Research on event sustainability has tended to focus on travel, energy, and spectator waste, which typically dominate overall carbon footprints (Bae & McCullough, 2025), while the procurement of branded goods has received comparatively little attention. This study addresses that gap by treating hat distribution as a bounded, quantifiable product system assessable through LCA.

1.4 Service Life and Reuse

One of the most consistent findings in textile LCA research is that extending a product's service life reduces its environmental impact. Production impacts are fixed once an item is manufactured; spreading them over more uses lowers the per-use burden. Sandin and Peters (2018), in a review of 41 published studies on textile reuse and recycling, found broad agreement that reuse delivers greater environmental benefits than recycling, and that both outperform landfilling or incineration. The key variable is the replacement rate: the degree to which a reused item actually prevents a new one from being produced. A more recent review of circular economy practices in textiles confirmed that extending service life consistently ranks as the highest-priority intervention for reducing environmental impacts (Abagnato et al., 2024).

At the Biathlon World Cup in Östersund, informal reuse practices are already in place: returning volunteers who received a hat in a previous edition do not receive a new one, and the supplier stores leftover stock for subsequent events. These practices reflect an awareness of resource efficiency but are not systematically formalised or tracked. This study examines whether more deliberate strategies - a formalised volunteer bring-back scheme and a reduction in free public giveaways

- could meaningfully reduce the number of new hats procured per edition, and by how much.

It should be noted that use-phase intensity differs across distribution channels: a hat sold through merchandise is more likely to be used repeatedly than one given away at a stand. However, since all scenarios in this study compare procurement strategies rather than individual use patterns, and since use-phase assumptions are held constant across all scenarios, this variation does not affect the comparative conclusions. The use phase is therefore excluded from the quantitative model, consistent with ISO 14044 guidance (ISO, 2006b). The absolute per-use environmental impact of each channel would require separate modelling with empirical use frequency data.

1.5 Life Cycle Assessment: Method and Application

Life Cycle Assessment is a standardised method for quantifying the environmental impacts of a product or service across its full life cycle, from raw material extraction to end-of-life treatment. It is governed by two ISO standards: ISO 14040, which sets out the general framework and principles, and ISO 14044, which specifies the requirements for conducting a study (ISO, 2006a; ISO, 2006b).

The functional unit is a central element of any LCA, as it defines what is being measured and compared. In textile research, functional units have been defined in various ways: per kilogram of fibre, per garment, or per use occasion. This variation makes cross-study comparisons difficult (Dhiwar & Bedarkar, 2025). In this study, the functional unit is one event edition of hat provisioning at the Biathlon World Cup in Östersund, corresponding to the full quantity of hats distributed across all three channels in a single edition. This reflects how the event organiser makes procurement decisions, as quantities are determined on a per-edition basis.

Scenario-based LCA is a well-established approach for evaluating circular economy interventions. Rather than describing only the current system, it models outcomes under alternative configurations and compares them on a consistent basis. This approach is used in both Sandin and Peters (2018) and Abagnato et al. (2024) in the textile reuse literature, and is applied here for the same reasons. The specific

modelling choices - software, database, impact method, and scenario definitions - are described in Section 2.

1.6 Aim and Research Question

The aim of this study is to quantify the climate change impact of hats distributed at the Biathlon World Cup in Östersund per event edition, and to evaluate how alternative distribution and procurement strategies could reduce those impacts. The International Biathlon Union has adopted climate targets for World Cup events, providing an organisational context in which quantified procurement-level interventions are directly relevant (IBU, 2020).

The research question is:

> What is the climate change impact of hats distributed at the Biathlon World Cup in Östersund per event edition, measured in kg CO_2e, and by how much could strategies such as a formalised volunteer bring-back scheme and reduced public giveaways reduce this impact?

2. Method

2.1 LCA Framework and Software

This study follows the ISO 14040/14044 framework for life cycle assessment, comprising four phases: goal and scope definition, life cycle inventory analysis, life cycle impact assessment, and interpretation (ISO, 2006a; ISO, 2006b). The modelling was carried out in SimaPro 9.1 (PRé Sustainability, 2021) using the ecoinvent 3 database with allocation, cut-off by classification as the primary background database. Industry data 2.0 was included as a supplementary library. Environmental impacts were characterised using the ReCiPe 2016 Midpoint (H) method. Climate change, expressed in kg CO_2e, is reported as the primary impact category, consistent with standard practice in textile LCA research (Dhiwar & Bedarkar, 2025). Additional midpoint categories were calculated but are not reported, as the study goal and available primary data are insufficient to support robust conclusions across a wider set of categories; this limitation is discussed in Section 4.3.

2.2 Functional Unit and System Boundary

The functional unit was defined as one event edition of hat provisioning at the Biathlon World Cup in Östersund, corresponding to the full quantity of hats distributed across three channels in a single edition: 1,000 hats issued to volunteers as part of their event equipment, 30 given away to the public at entertainment stands, and 500 sold through the merchandise shop. In total, this amounted to 1,530 hats per edition, with a combined weight of 154.5 kg based on a measured unit weight of 101 g per hat. This functional unit reflected how procurement decisions are made at the event, as hat quantities are determined on a per-edition basis.

The system boundary is cradle-to-grave, starting from raw material extraction and ending with end-of-life treatment, with manufacturing, transport, and distribution in between. Excluded from the system boundary are: the use phase, event infrastructure, spectator and volunteer travel, catering, and all other merchandise categories. These exclusions reflect the study scope, which is limited to hats as a bounded, decision-relevant product system. The full system boundary and distribution flows are illustrated in Figure 1.

The use phase was excluded from the quantitative model because all scenarios in this study share identical use-phase assumptions, and exclusion therefore does not affect comparative conclusions between scenarios. This is consistent with ISO 14044 guidance that life cycle stages identical across all alternatives may be excluded when they do not influence the conclusions (ISO, 2006b).

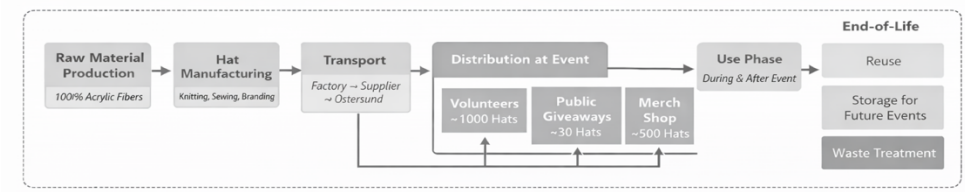


Figure 1: *System boundary diagram showing the cradle-to-grave scope and the three distribution channels*

2.3 Scenarios

Four scenarios were modelled on the basis of the same functional unit. The baseline scenario (S0) represented the current procurement system, in which all hats were newly produced each edition. S1

modelled a formalised volunteer bring-back scheme in which 30% of volunteers returned their hat at the end of the event for redistribution the following year, reducing new hat procurement in the volunteer channel from 1,000 to 700. S2 modelled a reduction in public giveaways from 30 to 10 units. S3 combined both interventions simultaneously. The scenario definitions and resulting quantities are summarised in Table 1.

Table 1: Scenario definitions and total hat quantities per event edition.

Scenario	Description	Total hats	Total weight (kg)
S0 Baseline	Current system - full procurement each edition	1,530	154.5
S1 Reuse	30% volunteer bring-back, 300 fewer new hats procured	1,230	124.2
S2 Reduced giveaways	Public giveaways reduced from 30 to 10; improved demand planning	1,510	152.5
S3 Combined	S1 and S2 combined	1,210	122.2

The 30% reuse rate applied in S1 was selected as a central estimate for sensitivity testing, reflecting an informal bring-back practice already in place at the event and considered operationally feasible by the event organiser. It does not represent a validated return rate from comparable schemes. The sensitivity of results to this assumption is tested across a range of 10–50% in Section 2.5.

2.4 Life Cycle Inventory

Each life cycle stage was modelled as a separate unit process in SimaPro. The inventory covers four stages: raw material production, manufacturing, transport, and end-of-life treatment. All hats are 100% acrylic with a measured unit weight of 101 g, confirmed by physical weighing of a sample provided by the event organiser. The datasets used for each stage are summarised in Table 2 in Section 3.1.

A dedicated acrylic fibre dataset was not available in the version of ecoinvent accessed through the institutional licence. Polyester fibre (RoW) was therefore used as a proxy for the raw material stage. This substitution is supported by van der Velden et al. (2014), who report

cradle-to-gate carbon footprints of 3.04 kg CO_2e per kg for acrylic and 2.70 kg CO_2e per kg for polyester (PET) at the raw material stage, indicating broadly similar production-stage profiles. However, acrylic production involves the polymerisation of acrylonitrile, a petrochemical monomer with a distinct chemical pathway from polyester, and may carry higher climate impacts under some characterisation approaches. The proxy therefore represents a potentially conservative estimate of baseline impacts. Absolute values reported in this study should be interpreted with this uncertainty in mind. The direction of scenario comparisons is not affected, as the same proxy applies equally across all scenarios.

Manufacturing electricity was modelled using the Chinese national grid (market group, CN), consistent with the manufacturing origin indicated on the hat label. The electricity input per hat was estimated at 0.1 kWh per hat. Van der Velden et al. (2014) report knitting electricity of approximately 0.14 kWh per kg of fabric for a mid-range yarn (300 dtex), equivalent to around 0.014 kWh for a 0.101 kg hat; the higher value of 0.1 kWh per hat was adopted as a conservative allowance covering making-up and handling in addition to knitting. Because this input is uncertain, its influence was tested across a range of 0.05–0.2 kWh per hat in the sensitivity analysis (Section 2.5).

Sea freight was estimated at approximately 20 000 km, corresponding to the container shipping route from China to Gothenburg via the Suez Canal. Road transport from Gothenburg to Östersund was estimated at approximately 1,500 km. These distances were derived from standard routing estimates for container freight and road haulage between these origin and destination points. End-of-life was modelled as municipal incineration of soiled waste textile using a RoW dataset, reflecting the predominant treatment pathway for non-recyclable synthetic textiles in Sweden (Sandin & Peters, 2018). Sensitivity to this assumption is tested in Section 2.5.

The study used only aggregate procurement data provided by the event organiser with consent and involved no human subjects or personal data, so no ethical review was required; all data sources and assumptions are reported transparently in line with ISO 14044 (ISO, 2006b).

2.5 Sensitivity Analysis

Sensitivity analysis was conducted to test the influence of four key parameters on the climate change results: the reuse rate assumed in S1, the end-of-life treatment pathway, the hat unit weight, and the manufacturing electricity input. For the reuse rate, S1 was modelled at return rates of 10%, 30%, and 50%, corresponding to total procurement volumes of 1,430, 1,230, and 1,030 hats respectively. This range captures uncertainty in volunteer compliance and the operational feasibility of a bring-back scheme at this event scale.

For end-of-life treatment, the baseline assumption of municipal incineration was compared against landfill, using a waste textile landfill dataset (RoW) as the alternative. For hat unit weight, the measured baseline of 101 g was varied by ±20%, representing plausible variation in product specification across event editions or suppliers. The manufacturing electricity input was varied between 0.05 and 0.2 kWh per hat, covering a plausible range around this estimate. All sensitivity runs were calculated using the same impact method and impact category.

3. Results

3.1 Life cycle inventory

The compiled life cycle inventory for the functional unit is summarised in Table 2, with all inputs scaled to the per-hat values used in the model. The parameters and their sources are described in Section 2.4.

Table 2: Inventory datasets used for each life cycle stage.

Life cycle stage	Dataset	Geography	Amount per hat	Unit
Raw material	Fibre, polyester, production finished	RoW	0.101	kg
Manufacturing	Market group for electricity, medium voltage	CN	0.1	kWh
Sea transport	Transport, freight, sea, container ship, market	GLO	2.02	tkm
Road transport	Transport, freight, lorry, unspecified, market	RoW	0.1515	tkm
End of life	Waste textile, soiled, municipal incineration	RoW	0.101	kg

RoW = Rest of World; CN = China; GLO = Global; tkm = tonne-kilometre; kWh = kilowatt-hour

3.2 Baseline System (S0)

The baseline scenario (S0) represents the current procurement system at the Biathlon World Cup in Östersund, in which all 1,530 hats are newly produced each event edition. The total climate change impact of the baseline system was 1,670 kg CO_2e per event edition. This corresponds to approximately 1.09 kg CO_2e per hat across the full cradle-to-grave system boundary.

The volunteer channel comprised the largest procurement volume, at 1,000 of the 1,530 hats (65%), followed by merchandise (500 hats, 33%) and giveaways (30 hats, 2%). Each channel's share of total impact corresponded to its share of procurement volume.

3.3 Scenario Comparison

The results of all four scenarios are presented in Table 3 and illustrated in Figure 2. Reductions relative to the baseline ranged from 2% (S2) to 21% (S3).

Table 3: Climate change results for all scenarios, expressed in kg CO_2e per event edition.

Scenario	Description	kg CO_2e	vs S0 (%)
S0 Baseline	Current system - full procurement each edition	1,670	-
S1 Reuse 30%	30% volunteer bring-back, 300 fewer new hats	1,350	-19%
S2 Reduced giveaways	Public giveaways reduced from 30 to 10	1,640	-2%
S3 Combined	S1 and S2 combined	1,320	-21%

Relative to the baseline of 1,670 kg CO_2e, S1 produced a total impact of 1,350 kg CO_2e, a reduction of 320 kg CO_2e (19%), corresponding to the avoided production of 300 hats in the volunteer channel. S2 produced 1,640 kg CO_2e, a reduction of 30 kg CO_2e (2%), corresponding to the reduced procurement in the giveaway channel. S3 produced 1,320 kg CO_2e, a reduction of 350 kg CO_2e (21%); within S3, the reduction attributable to S2 beyond S1 was 30 kg CO_2e. Figure 2 illustrates these results.

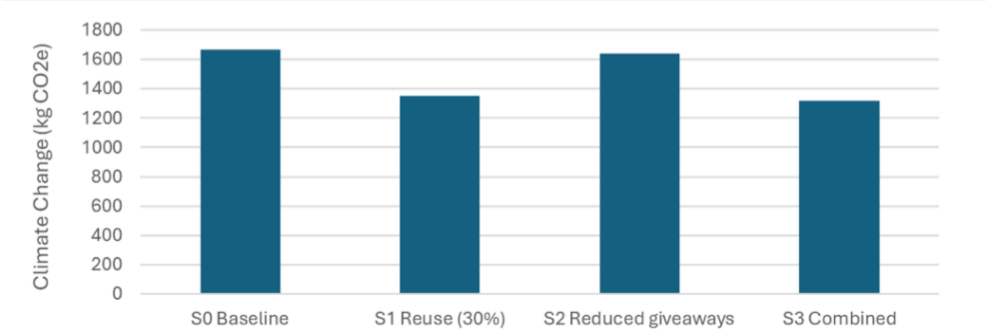


Figure 2: Climate change results (kg CO_2e per event edition) for the four procurement scenarios, illustrating the dominant contribution of the volunteer reuse intervention (S1) relative to reduced giveaways (S2).

To illustrate the cumulative effect of procurement decisions over a planning horizon, Figure 3 projects the climate change impact of hat distribution at Östersund over five editions under three procurement profiles. The baseline (1,530 hats annually) accumulates 8,350 kg CO_2e over this period. A 30% volunteer reuse scenario (approximately 1,230 newly procured hats annually) accumulates 6,750 kg CO_2e, a saving of 1,600 kg CO_2e. A 50% reuse rate would yield a cumulative saving of 2,700 kg CO_2e over the same period. These are linear projections assuming stable event size and consistent scheme performance, but they demonstrate how a single procurement intervention compounds over time.

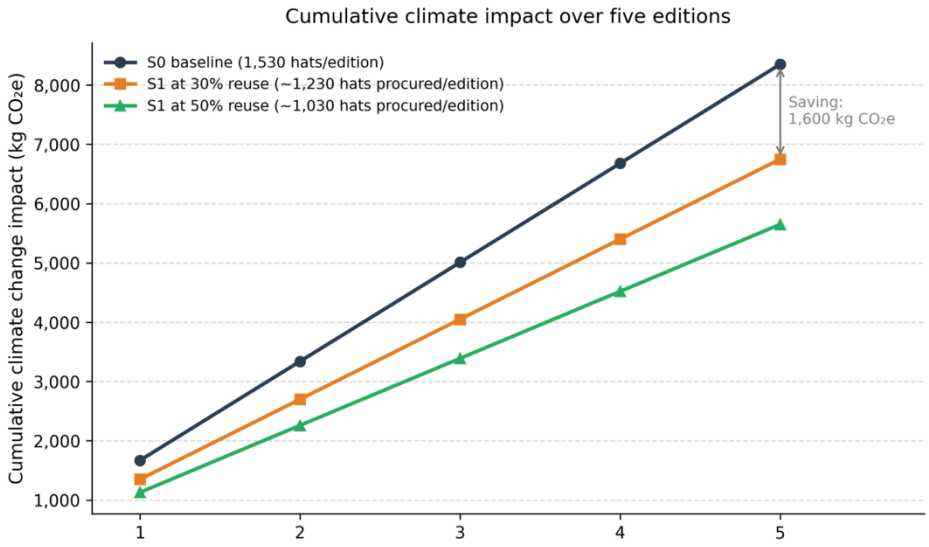


Figure 3: Cumulative climate change impact (kg CO_2e) of hat procurement at the Biathlon World Cup in Östersund over five editions, comparing the baseline (S0) with volunteer reuse scenarios at 30% and 50% return rates.

3.4 Sensitivity Analysis

Sensitivity analysis was conducted for four parameters: the volunteer reuse rate, the end-of-life treatment assumption, the hat unit weight, and the manufacturing electricity input. Results are presented in Tables 4-7 respectively.

3.4.1 Reuse rate

The reuse rate was the most consequential parameter tested. As shown in Table 4, varying the volunteer return rate between 10% and 50% produced a total impact range of 1,130 to 1,560 kg CO_2e, representing reductions of 7% to 32% relative to S0. Even at a low return rate of 10% (100 returned hats), the scheme delivered a 7% reduction. At 50% (500 returned hats), the reduction reached 32%. The linear relationship between return rate and impact reduction is expected, as each returned hat directly substituted one newly produced hat on a one-to-one basis.

Table 4: Sensitivity of S1 results to the assumed volunteer reuse rate.

Reuse rate	Total hats procured	kg CO_2e	vs S0 (%)
10%	1,430	1,560	-7%
30% (base case)	1,230	1,350	-19%
50%	1,030	1,130	-32%

3.4.2 End-of-life treatment

The choice of end-of-life treatment had a negligible effect on results. As shown in Table 5, replacing municipal incineration with landfill increased total baseline impact from 1,670 to 1,750 kg CO_2e, a difference of 80 kg CO_2e or 5%. End-of-life was therefore confirmed as a minor driver of climate change impact for this product system.

Table 5: Sensitivity of S0 baseline to end-of-life treatment assumption.

End-of-life assumption	kg CO_2e	vs S0 incineration (%)
Municipal incineration (base case)	1,670	-
Landfill	1,750	+5%

3.4.3 Hat unit weight

Hat unit weight scaled results proportionally, as shown in Table 6. A 20% reduction in weight (81 g) reduced baseline impact by 20% to 1,340 kg CO_2e, while a 20% increase (121 g) raised it by 20% to 2,000 kg CO_2e. This strictly linear relationship confirmed that total impact was directly proportional to mass, since all upstream inventory inputs were scaled by unit weight.

Table 6. Sensitivity of S0 baseline to hat unit weight (±20% variation from 101 g).

Unit weight	kg CO_2e	vs S0 base case (%)
81 g (-20%)	1,340	-20%
101 g (base case)	1,670	-
121 g (+20%)	2,000	+20%

3.4.4 Manufacturing electricity input

The electricity input per hat was tested across the range reported in the literature for knitting and assembly of simple knitted accessories. As shown in Table 7, varying the input between 0.05 and 0.2 kWh per

hat produced a total impact range of 1,590 to 1,830 kg CO_2e, a variation of −5% to +10% relative to the base case. This confirmed that the electricity input assumption had a minor influence on results. The raw material stage, rather than manufacturing energy, determined the majority of climate change impact.

Table 7. Sensitivity of S0 baseline to manufacturing electricity input.

Electricity input	kWh per hat	kg CO_2e	vs S0 (%)
Low	0.05	1,590	-5%
Base case	0.1	1,670	-
High	0.2	1,830	+10%

Across the four parameters tested, the reuse rate produced the widest output range, from 1,130 to 1,560 kg CO2e. End-of-life treatment changed the baseline by 5%, electricity input by -5% to +10%, and unit weight scaled it linearly by ±20%. The scenario ranking was unchanged across all parameter ranges tested.

4. Discussion

4.1 Which Lever Matters and Why

The central finding of this study is that volunteer hat reuse is the only intervention among those modelled that produces a materially significant reduction in climate change impact. S1 delivered a 19% reduction relative to the baseline; S2 delivered 2%; S3, combining both, added only two percentage points beyond S1 alone. This pattern reflects a structural feature of the system rather than a coincidence of the specific values chosen.

The giveaway channel represents less than 2% of total procurement volume. No amount of demand planning focused exclusively on that channel can produce a meaningful environmental outcome at the system level. The result of S2 should therefore be read as a diagnostic: it reveals where the system's volume is concentrated and, by implication, where reduction strategies must be directed. The volunteer channel, at 65% of total hats, is where procurement decisions carry environmental weight.

The logic underlying these results is well established in the textile LCA literature. Sandin and Peters (2018) demonstrate, in a review of 41 studies, that reuse consistently outperforms recycling and end-of-life management as a strategy for reducing environmental impact, because it avoids the production of new items entirely. The critical variable in reuse modelling is the substitution rate: the degree to which a returned item actually prevents a new one from being produced. In this study, a one-to-one substitution was assumed, meaning each returned hat displaced exactly one new hat in the following edition. This is a conservative and transparent convention; the conditions under which it holds in practice are discussed in Section 4.3.

4.2 Comparison with Existing Literature

The baseline impact of approximately 1.09 kg CO_2e per hat is consistent with published benchmarks for synthetic knitted accessories. Van der Velden et al. (2014) report cradle-to-grave climate change impacts for acrylic and polyester textiles in a range broadly compatible with the per-unit result here, noting that production-stage impacts dominate and that end-of-life contributes minimally for synthetic fibres. The near-zero sensitivity to end-of-life treatment found in this study (+5% for landfill versus incineration) is consistent with that finding.

The scenario reductions also align with the general magnitude of reuse benefits reported in the broader textile reuse literature. Abagnato et al. (2024), in a review of LCA applications to textile circular practices, report that service life extension consistently yields the highest environmental benefit among circular strategies, often in the range of 20–40% reduction in climate impact depending on baseline service life and replacement rate. The 19% reduction found in S1 at a 30% return rate falls within this range and scales predictably: at 50%, the 32% reduction approaches the upper bound. This correspondence strengthens confidence in the model and confirms that the order of magnitude is plausible.

A finding with less precedent in the existing literature is the structural irrelevance of the giveaway channel. Most textile LCA studies focus on product categories with fixed, single distribution pathways and do not need to account for multi-channel procurement with highly unequal channel volumes. This study demonstrates that when procurement is

segmented across channels of very different scales, the environmental relevance of each channel is determined primarily by its volume share. The implication for event sustainability research is that reducing or eliminating small-scale giveaways may serve communication or branding purposes but should not be presented as environmentally significant without first establishing whether the channel represents a material share of total procurement.

The study also contributes to a literature gap identified by Dhiwar and Bedarkar (2025), who note that event-related textile distribution has received comparatively little attention in peer-reviewed LCA research. The functional unit developed here - one event edition of hat provisioning - offers a replicable framing for future studies of this type.

4.3 Limitations and Uncertainties

Several limitations affect the interpretation of this study's results, grouped below by type.

Inventory data. The most significant inventory limitation is the use of a polyester fibre dataset as a proxy for acrylic, due to the absence of a dedicated acrylic dataset in the version of ecoinvent accessible through the institutional licence. This substitution may underestimate baseline impacts, since acrylic production involves acrylonitrile polymerisation, which can carry higher characterised impacts than polyester under some methods. The direction and ranking of scenario conclusions are not affected, but absolute values should be treated as potentially conservative estimates.

Substitution and reuse modelling. The one-to-one substitution assumption treats every returned hat as directly preventing the production of one new hat. This may overestimate benefits if returned hats are reissued to recipients who would not have received a new hat regardless - for example, returning volunteers who already own a hat from a previous edition and would not have been included in the following year's procurement. Conversely, it may underestimate benefits if returned hats are used across multiple subsequent editions. The model does not account for quality degradation over time, which could limit the number of reuse cycles achievable in practice.

Edition-specific branding. Event-branded textiles frequently carry year designations, edition-specific sponsor logos, or graphics tied to a

particular championship cycle. The substitution logic in S1 assumes that returned hats from one edition can be reissued in the next. If branding changes annually, the practical reuse pool may be smaller than the model implies. A move toward event-series branding rather than year-specific designations would substantially expand the reuse window.

Empirical reuse data. The reuse rates tested in the sensitivity analysis (10–50%) are estimated ranges rather than empirically derived values. The 30% base-case assumption should be treated as a modelling choice rather than a validated behavioural prediction. Comparable contexts - deposit-return schemes, library lending systems, and equipment-rental models - suggest that return rates depend heavily on scheme design, with clear communication, convenient return points, and visible environmental rationale as key determinants.

Single impact category. This study reports climate change as the primary and only fully discussed impact category, consistent with standard practice in textile LCA research (Dhiwar & Bedarkar, 2025). Additional midpoint categories were calculated in SimaPro but are not reported, because the polyester proxy introduces greater uncertainty in toxicity-related categories than in climate change, and because climate change is the most established and communicable indicator for organiser decision-making. Future work should consider whether a broader impact profile alters the scenario ranking.

Single-event case study. The study is limited to one product type (hats), one material (100% acrylic), and one event (Biathlon World Cup in Östersund). The conclusions are directly applicable to this system but cannot be generalised to other merchandise categories, fibre compositions, or events with different distribution structures without additional case-specific modelling. Events with larger giveaway volumes relative to total procurement might find the S2-type intervention materially more significant than found here.

4.4 Broader Implications

This study contributes to a broader discussion about the role of avoided production in reducing the environmental impact of textiles. The dominant narrative in sustainable fashion and event sustainability

has historically emphasised recycling, material substitution, and end-of-life management. The results presented here reinforce what the reuse literature has argued for some time: these downstream interventions are structurally less effective than reducing the number of items produced (Sandin & Peters, 2018; Abagnato et al., 2024). For acrylic textiles specifically, where neither biodegradation nor commercially viable recycling is available, the only effective circular strategies are those that extend the useful life of items already in circulation.

Events often frame their sustainability efforts in terms of material choices - switching from conventional to recycled-content textiles or from synthetic to natural fibres - without quantifying whether such substitutions actually reduce total production volume or simply change the composition of an unchanged quantity. This study demonstrates that procurement volume, rather than material composition, is the primary driver of climate impact for this type of product system. A shift from virgin to recycled acrylic would reduce the raw material production burden but would leave 1,530 units produced per edition. A bring-back scheme returning 300 hats to circulation eliminates those 300 units from the next edition's procurement entirely.

More broadly, the per-edition functional unit and multi-channel system framing developed here offer a practical template for other events and organisations distributing branded textiles at scale. The IBU Biathlon World Cup series visits eight to nine venues annually, and applying the same modelling framework across venues would allow for network-level procurement optimisation. As shown in Figure 3 in Section 3.3, a single procurement intervention compounds meaningfully over time: even a 30% return rate accumulates a saving of 1,600 kg CO_2e over five editions. The Östersund case therefore functions as a methodological template whose proportions and trade-offs are likely to apply at substantially larger scales, where the absolute environmental gains from procurement reform would be correspondingly greater.

4.5 Practical Recommendations for the Organiser

The following recommendations are presented in order of estimated environmental impact. Each is directly grounded in the model results rather than derived from general sustainability principles.

First, formalise the volunteer bring-back scheme and set a target return rate. The current informal practice of storing leftover hats shows awareness of resource efficiency, but without a formalised mechanism, return rates are unpredictable and uncounted. A dedicated collection point at the event close, combined with clear communication to volunteers about the scheme's environmental rationale, is the most impactful operational change available. A 30% return rate would reduce procurement impact by 19%; each additional ten percentage points yields a further 6–7% reduction. A target of 40–50% is a reasonable ambition for a well-communicated scheme.

Second, link the bring-back scheme directly to procurement planning. The environmental benefit of returned hats is only realised if procurement in subsequent editions is reduced by an equivalent number. The organiser should establish a formal process in which the quantity of returned and inspected hats is subtracted from the following edition's procurement order before that order is placed.

Third, track volunteer hat distribution by individual, not only by edition. Maintaining a simple record of which volunteers received hats in which editions would allow the organiser to quantify effective reuse already occurring in the system and identify the actual net procurement need more precisely. This data would also improve the accuracy of future LCA modelling.

Fourth, consider the material specification of future hat procurement. The sensitivity analysis shows that a 20% reduction in unit weight yields a 20% reduction in total impact, equivalent in magnitude to the S1 scenario at 30% return rate. If a new hat design is being sourced, requesting a lighter specification - where this does not compromise warmth and durability - would yield comparable environmental benefits with no operational change at the event itself. Weight-reduction and reuse strategies should be evaluated jointly, as a thinner knit may reduce the number of viable reuse cycles.

Fifth, treat the giveaway channel as a low priority for environmental reduction but a relevant target for waste prevention. The 30 publicly distributed hats are negligible at the system level, but making giveaways available only on request rather than distributing them proactively would reduce the risk of producing items that are not used.

4.6 Conclusion

This study quantified the climate change impact of hats distributed at the Biathlon World Cup in Östersund per event edition and evaluated three impact-reduction strategies using comparative scenario-based LCA. The baseline system, in which 1,530 acrylic hats are newly procured each edition, generated a climate change impact of 1,670 kg CO_2e. A formalised volunteer bring-back scheme at a 30% return rate reduced this to 1,350 kg CO_2e, a 19% reduction. Reducing public giveaways from 30 to 10 units reduced impact by 2%. The combined strategy yielded 21%, confirming that reuse is the dominant lever and giveaway reduction adds negligible marginal benefit at current channel volumes.

Sensitivity analysis showed that the bring-back scheme's performance depends critically on the return rate achieved: the range of 10–50% produced outcomes between −7% and −32% relative to the baseline, giving the organiser a quantified basis for setting a target. End-of-life treatment and unit weight variation were secondary sources of uncertainty and did not alter the scenario ranking.

The key practical conclusion is that environmental improvement in this system is achievable through operational procurement decisions rather than material substitution or end-of-life management. Formalising a bring-back scheme, linking returned hat inventories directly to procurement orders, and tracking volunteer hat distribution across editions are actions within the organiser's immediate control. The study's limitations - particularly the polyester proxy for acrylic, the absence of empirically derived reuse rates, and the single impact category reported - mean that absolute values carry uncertainty and should be treated as order-of-magnitude estimates. The comparative conclusions between scenarios are robust to these limitations, as they depend on the relative structure of the system rather than on specific inventory values.

Future work should collect empirical return rate data from comparable bring-back schemes, assess whether a broader set of impact categories alters the scenario ranking, and apply the per-edition modelling framework to other venues in the Biathlon World Cup series to identify network-level procurement optimisation opportunities.

References

Abagnato, S., Rigamonti, L., & Grosso, M. (2024). Life cycle assessment applications to reuse, recycling and circular practices for textiles: A review. Waste Management, 182, 74–90. https://doi.org/10.1016/j.wasman.2024.04.016

Bae, Y., & McCullough, B. P. (2025). Environmental impact and carbon emissions of sport events: The significance of Scope 3. WIREs Climate Change, 16(4), e70017. https://doi.org/10.1002/wcc.70017

Dhiwar, K., & Bedarkar, M. (2025). Life cycle assessment in fashion industry: A systematic review. Discover Sustainability, 6, 1214. https://doi.org/10.1007/s43621-025-02050-7

IBU (International Biathlon Union). (2020). IBU sustainability strategy 2020–2030. International Biathlon Union. https://assets.ctfassets.net/cz0vl36hcq0x/H8bDv652rkK8Z6UiMXYmpQ/c69aad9588c74f642714a3de1a6f9c2e/IBU_Sustainability_Strategy_2020_2030_v02.pdf

ISO. (2006a). ISO 14040: Environmental management – Life cycle assessment – Principles and framework. International Organization for Standardization.

ISO. (2006b). ISO 14044: Environmental management – Life cycle assessment – Requirements and guidelines. International Organization for Standardization.

Niinimäki, K., Peters, G., Dahlbo, H., Perry, P., Rissanen, T., & Gwilt, A. (2020). The environmental price of fast fashion. Nature Reviews Earth & Environment, 1(4), 189–200. https://doi.org/10.1038/s43017-020-0039-9

PRé Sustainability. (2021). SimaPro 9.1 [Software]. PRé Sustainability B.V. https://simapro.com

Sandin, G., & Peters, G. M. (2018). Environmental impact of textile reuse and recycling – A review. Journal of Cleaner Production, 184, 353–365. https://doi.org/10.1016/j.jclepro.2018.02.266

van der Velden, N. M., Patel, M. K., & Vogtländer, J. G. (2014). LCA benchmarking study on textiles made of cotton, polyester, nylon, acryl, or elastane. The International Journal of Life Cycle Assessment, 19(2), 331–356. https://doi.org/10.1007/s11367-013-0626-9

Terms of publishing

1. Mid Sweden University collaborates with DiVA (Digitala Vetenskapliga Arkivet).

DiVA is a digital repository for the registration and storage of research publications and independent works, and in DiVA, the documents can be made public.

2. By uploading and publishing the author's works in DiVA, these are made available to the public.

The expression "accessible to the public" means that virtually everyone has the opportunity to read the work. In this sense, the scientific community is seen as "the public". The author retains the copyright, and the public's use of the information in DiVA is regulated by the Copyright Act (See also bullet 9).

3. It is the responsibility of the author to hold the required copyright to publish the work in DiVA.

Since the author does not transfer any part of the copyright, the university/DiVA is not responsible for any violation of copyright rules regarding the author's work. The university only provides the platform, which means that the author is "the one who publishes" in DiVA.

4. It is the responsibility of the examiner to send the approved work, in full text, to the relevant faculty administrator.

It is the responsibility of the author to make sure that the work has been approved for publication; theses, degree projects and the like must be approved for publication before they are posted in DiVA. The faculty administrator is responsible for uploading and publishing the approved full text in DiVA.

5. The publication in DiVA rests on non-commercial grounds.

The university does not charge the author any fees for the publication in DiVA. The author is not entitled to financial compensation from the university for publishing in DiVA. The university does not have the right to charge fees for the public's use of the author's work in DiVA.

6. The university has the right to remove the author's work from DiVA if the author violates the terms of publishing.

According to the university's instructions for publishing in DiVA, the author is obliged to read and approve the Terms and Conditions of Publishing. These is confirmed by pressing a button in DiVA's registration module when uploading full text file.

7. The person who posts the full text in DiVA is responsible for ensuring that all authors of the work have been informed about and approved to the terms of publishing.

This regulates the responsibilities in multi-author works and such cases where someone else than the author, on their behalf, publishes the work in DiVA.

8. The author has the opportunity to refrain from part of his or her rights to the work.

By adding a special license to the work, such as Creative Commons, the authors can grant users the right to use the work within a wider framework than in accordance with the Copyright Act.

9. The terms of publishing apply when applicable, even if the university should transfer to a system solution other than DiVA.

Metadata and uploaded files should then be transferred to the new system.